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The Dissipative Structure of Value: Information Geometry and the Thermodynamics of NYC Real Estate

On Modelling with Madelung Flow Regression
A Research Report

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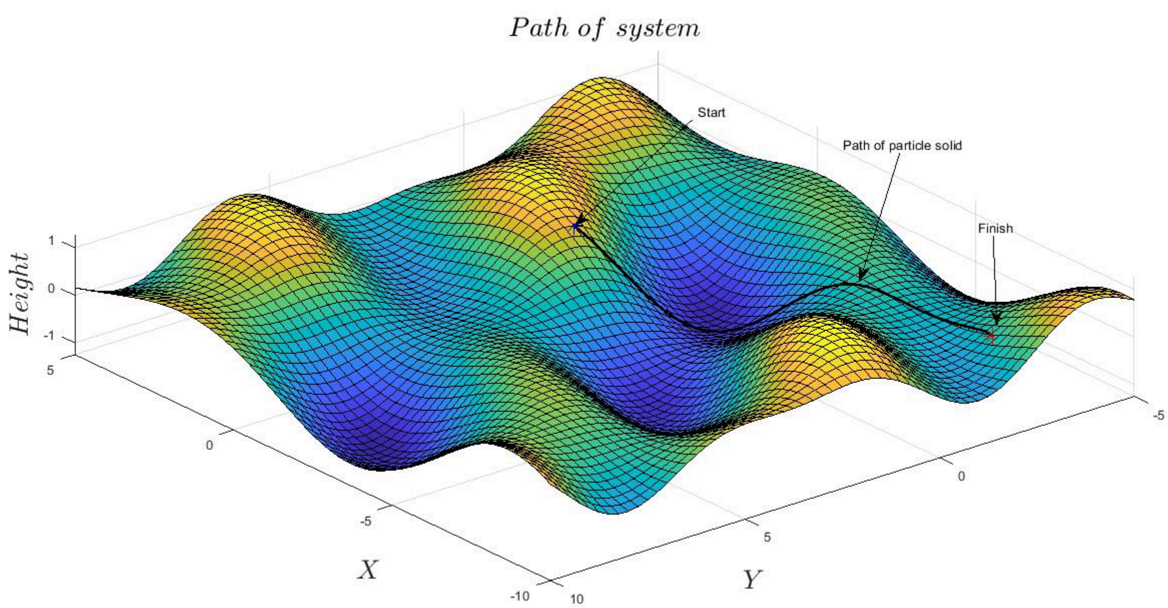

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Abstract:
This work develops a non-parametric, information-geometric framework for modeling the dynamics of complex systems, motivated by analogies to quantum hydrodynamics among other fields of physics and machine learning. The goal in this is the construction of an empirical effective field theory capable of modeling almost any kind of dynamical system of somewhat stochastic nature. By utilizing continuous-time kernel density estimation and score-based geometric constructions, the framework infers probability flows and dissipative structures directly from high-dimensional data without assuming parametric functional forms, allowing for a new kind of Causal World Modelling. We introduce a principled, information-theoretic scale selection procedure that balances estimator variance against Fisher information, mirroring renormalization-group reasoning.

The physical formalism (e.g., flow, dissipation, uncertainty) is employed as an interpretive analogy to characterize the informational topology of the system rather than as a claim of literal physical laws. The resulting framework is an effective, representation-dependent model that evolves with data resolution. As this represents an ongoing effort to bridge physical formalism with economic data science, the current nomenclature and interpretive framework are subject to iterative refinement.

Applied to U.S. housing market data, the framework identifies three distinct structural regularities: a characteristic temporal coherence scale, a robust informational constraint analogous to uncertainty relations in stochastic systems, and a dissipative component of inferred dynamics associated with structural maintenance. This application serves as an example of how quantum-inspired algorithmic frameworks can recover dynamically meaningful structure in complex, non-equilibrium datasets. *Git hub repository can be found near the bottom of this report.*

Introduction:
In August 2025 David Krakauer, president of the Santa Fe Institute, the world's leading research center on the study of complex systems said this on Neil deGrasse Tyson's *Star Talk* podcast:

"Warren Weaver, who was a mathematician, wanted to classify all of regularity or irregularity in the universe. He said there are simple phenomena; classical physics. It doesn't mean it's easy to understand, but it's simple. There are beautiful laws, elegant mathematical formalisms that explain it.

Then there's the world of what he called disorganized complexity. That's the study of gases, what we would now study with thermodynamics and statistical mechanics; irregular things that have beautiful descriptions.

And then in the middle, there's organized complexity. It's not a gas. It's an organism. It's an ant. It's a brain. It's a city. And in that space, what we realized really at the end of the 19th century is that we don't have good theories for it. We can do irregularity beautifully. We can do simplicity beautifully. But then you move to everything that we actually care about, in some sense as human beings, as animals, as living beings, and we kind of under-theorize it."

In many ways this describes my line of inquiry, the broad understanding of complexity through a novel framework descended from information theory, information geometry, thermodynamics, econometrics, and quantum hydrodynamics. Information theory, having originated from Claude Shannon, is the field of mathematics of quantifying information itself as a measurable physical quantity, at its core seeking to answer; "How much "surprise" or "uncertainty" is contained in a signal, and how efficiently can we encode it?". Information Geometry is the study of probability distributions as physical shapes, a fusion between differential geometry (the calculus of curved spaces, called manifolds, that look flat when zoomed in on, like the Earth) and statistics. Thermodynamics being the study of energy and entropy. Econometrics as the general science of causal modelling through hypothesis testing and regressions. Quantum mechanics models the microscopic word of particles as waves of probability rather than discrete points, breaking the traditional determinism of classical physics. Core to this is Schordinger's equation describing particles as wave packets through complex numbers. Erwin Madelung in 1926, the same year Schrodinger published his equation, released a version of it rewritten in hydrodynamic variables, effectively describing the quantum world as a flowing probability fluid rather than Schrodinger's amorphous and hard to conceive complex valued wave packets. My work combines these fields, often siloed into one continuous field-like theory, fields being a mathematical object that encodes mathematical properties over a space/manifold. When applied to economic data, it constitutes a new theory of the economy based on these principles. 2025 marks the 100th anniversary of the study of quantum mechanics. It excites me to present contributions related to this that can be applied to macro phenomena.

The Data:

To establish a comprehensive dataset of New York City property transactions, I developed a custom automated web crawler to harvest historical Rolling Sales data directly from the NYC Department of Finance (Link: Rolling Sales Data). The script utilized a "shotgun" request pattern to handle inconsistent URL structures and file formats (both .xls and .xlsx) across the target period of 2015 to the end of 2024. This process iteratively retrieved annualized sales files for all five boroughs, standardizing them into a unified structure despite varying header formats over the decade.

Once acquired, I subjected the raw data to a rigorous cleaning and harmonization pipeline to ensure the integrity of the thermodynamic variables. Non-numeric characters such as currency symbols and commas were stripped from the Price, BuildingArea, and LandArea fields, which I then coerced into numeric types. To isolate genuine market activity from non-arms-length transfers, such as familial deed transfers or administrative corrections, I applied a strict filter excluding records with a sale price below $1,000 or a gross building area under 100 square feet. I subsequently concatenated the cleaned extracts into a single master dataframe, nyc_real_estate_history(all).csv, providing a continuous and granular field of valid market transactions for the subsequent geometric analysis. The data set was also adjusted for inflation (September 2025 dollars).

The Madelung Object Model

The cornerstone of this analysis is what I term the "Madelung object," a continuous-time kernel density estimator (KDE) that constructs a smooth, differentiable probability field over the space of log-price, log-building-size, and time. The naming reflects the mathematical utility of this object: just as Erwin Madelung reformulated the Schrödinger equation into hydrodynamic variables, expressing quantum mechanics as a probability fluid, this estimator reframes market dynamics as a fluid of transaction probability. The density ρ represents market liquidity, while its evolution encodes the changing preferences and "drift" of market actors.

Formalism of the Spatio-Temporal Field

Mathematically, the engine treats the market as a manifold where the density of transactions is governed by the local geometry of information. For any query time T, the algorithm estimates the joint probability density $\rho(\mathbf{x}, T)$ of observing a transaction at spatial coordinates $\mathbf{x} = (x, y)$. This is not a static histogram, but a temporally weighted estimation using a Product Kernel:

$$\rho(\mathbf{x}, T) = \frac{\sum_{i=1}^{n} w_i(T) \cdot \mathcal{K}_{\mathbf{H}}(\mathbf{x} - \mathbf{x}_i)}{\sum_{j=1}^{n} w_j(T)}$$

The Temporal Weighting (Strict Causal Constraint):

The weights $w_i(T)$ are defined by a Gaussian kernel with bandwidth σ_t. Crucially, I have implemented a Strict Causal Constraint to ensure the model respects the arrow of time for predictive analysis:

$$w_i(T) = \begin{cases} \exp\left(-\frac{(T-t_i)^2}{2\sigma_t^2}\right) & \text{if } t_i \leq T \\ 0 & \text{if } t_i > T \end{cases}$$

This makes the engine a recursive non-parametric filter. Information is only permitted to flow from the past into the present, allowing the probability distribution to evolve smoothly; "flowing" from one market regime to another without the jitter and look-ahead bias of discrete windowing.

Adaptive Geometry and the Local Metric:

The engine does not assume a Euclidean space. Instead, it estimates the local geometry of the market manifold at time t by calculating a Weighted Covariance Matrix Σ_t from the active historical subset. In the implementation, this is inverted to yield the Precision Matrix Σ_t^{-1}, which acts as the local Metric Tensor.

This tensor defines the "Informational Distance" (Mahalanobis distance) between market states. The covariance is scaled into a bandwidth matrix using the effective sample size ($N_{eff} = 1/\sum \tilde{w}_i^2$) via Scott's Rule:

$$\mathbf{H}_t = N_{eff}^{-\frac{1}{d+4}} \Sigma_t$$

This ensures the manifold contracts in high-liquidity regimes (high precision), capturing fine-grained microstructure, and expands in low-liquidity regimes to prevent the "evaporation" of the fluid into stochastic noise. Crucially, the Precision Matrix Σ_t^{-1} serves as the geometric "measuring stick" for the kernel, distorting the space so that a 10 dollar price move in a volatile regime is treated as statistically equivalent to a 1 dollar move in a stable regime.

Analytical Kinematics of the Velocity Field & th Conservation of Probability:

Because the KDE is constructed from Gaussian basis functions, we can compute its derivatives analytically. The implementation explicitly calculates the spatial gradients (∇P) and the time derivative ($\partial_t P$) at every point in the grid.

A critical innovation in this implementation is the correction for Normalization Drift. Since the temporal kernel allows the total weight of the system (W) to "breathe" as transactions enter and exit the window, the time derivative is computed using the exact quotient rule:

$$\frac{\partial P}{\partial t} = \frac{1}{W} \sum \dot{w}_i K_i - P\left(\frac{\dot{W}}{W}\right)$$

This second term corrects for the changing normalization constant, ensuring that the calculated evolution is due to market dynamics, not artifacts of the sliding window. From these variables, we derive the Velocity Field ($\mathbf{v}$), which is mathematically the Score Function of the distribution:

$$\mathbf{v}(x,t) = \nabla \ln P(x,t) = \frac{\nabla P}{P}$$

In this hydrodynamic framework, this vector represents the "Osmotic Drift" or Information Velocity—the effective force driving prices toward the regions of highest liquidity. By calculating the score field analytically, the engine allows for the downstream computation of the Fisher Information Metric, derived from the outer product of these velocity vectors ($\mathbf{v}\mathbf{v}^T$), which defines the global curvature of the market state space.

The Dissipative Continuity Equation (Fokker-Planck Inversion):

These raw kinematic variables allow us to invert the Fokker-Planck equation, bridging the gap between quantum-hydrodynamic-like flow and thermodynamic-like dissipation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = \mathcal{S}(\mathbf{x}, T)$$

In this formulation, $\mathcal{S}$ represents the Dissipative Source Term (or "Market Heat"). Instead of assuming a model and simulating forward, we take the observed ρ and its derivatives as the ground truth. We solve this equation to extract the hidden dynamic variables: the drift velocity ($\mathbf{v}$) captures systematic market behavior, how prices adjust and building heterogeneity evolves, while the diffusion (represented within the residual $\mathcal{S}$) quantifies market friction and microstructure uncertainty.

The elegance of this formulation is that it requires no traditional assumptions about demand curves, equilibrium conditions, or behavioral hypotheses. It describes what the market *actually does* through its revealed transaction pattern, letting the data speak through its own geometric structure.

The Data Manifold and Fiber Bundle Architecture:

Conceptually, the Madelung Object fundamentally redefines the structure of financial data. Rather than treating the market as a discrete time-series or a sequence of static histograms, this model constructs a Fiber Bundle Manifold. In this geometric framework, Time serves as the base manifold, while the spatial dimensions (log-price and log-size) form the fiber attached to every temporal point.

The kernel density estimator acts as the connection on this bundle, strictly defining how these spatial fibers are "stitched" together. By smoothing over the temporal dimension with a continuous kernel, the model transforms isolated transaction events (Dirac deltas) into a unified, differentiable surface. This continuity is non-trivial; it ensures that the tangent space is well-defined at every coordinate (p, s, t).

Consequently, we are not limited to discrete difference equations. We can perform rigorous calculus on the market structure itself. The gradient vectors (∇P) and time derivatives ($\partial_t P$) live within these tangent spaces, allowing us to compute the instantaneous velocity and acceleration of liquidity. This architecture effectively converts the jagged, stochastic reality of trade logs into a smooth Riemannian manifold, where the flow of information follows defined geometric geodesics.

Assumptions

To treat the New York City real estate market as a Madelung fluid, the model rests on three core statistical assumptions:

Continuum Approximation: While transaction data is discrete, the market state can be represented as a continuous manifold ρ given a sufficient sampling density σ_t.

Local Analyticity: Market transitions are sufficiently smooth at the chosen bandwidth to permit analytical differentiation.

Informational Sufficiency: The chosen state-space dimensions, log-price (y) and log-building-size (x), capture the essential degrees of freedom required to describe the system's "Value Potential."

Computational Implementation

To make this mathematically tractable over a dataset of this magnitude, I developed two parallel implementations of the estimator:

CPU Implementation (NumPy/SciPy):
For smaller subsets or rapid prototyping, this version utilizes scipy.stats.gaussian_kde with custom temporal weighting. It leverages standard linear algebra libraries to compute covariance matrices and handle the weighted kernel summation efficiently.

GPU Acceleration (PyTorch):
To analyze the full high-resolution field (100×100 spatial grids over 3,650 days), the computational cost of evaluating Gaussian kernels scales quadratically ($N_{grid} \times N_{data}$). I implemented a hardware-accelerated version using PyTorch that treats the KDE summation as a massive tensor operation. This allows the pairwise distance computations and exponentiations to be broadcast across thousands of CUDA cores simultaneously, reducing the computation time for the full 10-year "Madelung object" generation from hours to minutes.

For reference all of the computation times in this research are manageable without GPU assistance, with the exception of the bandwidth optimization that I am about to cover. The GPU assistance was specifically added because that calculation would take hours, maybe even more than a day to complete with the CPU alone.

On Eliminating Look-Ahead Bias, The Causal Kernel

A critical structural refinement was necessary to ensure the validity of predictive experiments. The standard Continuous-Time KDE utilizes a symmetric Gaussian kernel in the temporal dimension, which effectively smooths the manifold for geometric visualization by weighting contributions from both past and future transactions. While this bidirectional smoothing is ideal for mapping the static structure of the market phase space and descriptive analysis of dynamics, it introduces a fatal look-ahead bias for predictive analysis. If the density estimate at a given time includes information from future price shocks, the resulting metrics would implicitly contain the very signal they purport to predict, rendering any causality tests invalid.

To resolve this, the Madelung object was re-engineered to support a strict causal mode for the time-series analysis. This adaptation forces the estimator to respect the arrow of time by reconstructing the probability density at any historical date using only the transaction data available up to that specific moment. This distinction allows us to retain the high-fidelity geometric visualizations for structural analysis while guaranteeing the integrity of the econometric validation.

Bandwidth Optimization
The central challenge in constructing a continuous-time probability field is determining the optimal temporal bandwidth (σ_t). This parameter effectively sets the "shutter speed" of the observation. If σ_t is too short, the model captures transient microstructure noise (high variance); if too long, it oversmooths the rapid phase transitions characteristic of market regime shifts (high bias). Note that because the underlying price data was already inflation-adjusted, this smoothing concern pertains to structural volatility rather than monetary inflation itself.

To rigorously select this parameter, I developed an optimization framework rooted in Information Geometry. Rather than relying on heuristic rules of thumb, I sought the bandwidth that maximized the structural resolution of the phase space, specifically, the clarity of the relationship between Market Liquidity (ρ) and Kinematic Volatility ($|\mathbf{v}|$).

This problem is governed by the **Gabor Uncertainty Principle**, the information-theoretic analogue to Heisenberg's Uncertainty Principle in quantum mechanics ($\Delta x \Delta p \geq \hbar/2$). Just as one cannot simultaneously know a particle's position (x) and momentum (p) with infinite precision, in market dynamics, we face a fundamental rigid lower bound on the product of temporal resolution (Δt) and frequency resolution ($\Delta \omega$):

$$\Delta t \cdot \Delta \omega \geq \frac{1}{4\pi}$$

In the context of the Madelung Object, this manifests as a collision between two opposing statistical limits:

1. The Microscopic Limit ($\sigma_t \to 0$): We attempt to capture instantaneous velocity ("momentum"), but the effective sample size N_{eff} vanishes. Governed by the Cramér-Rao Bound ($Var(\hat{\theta}) \geq I(\theta)^{-1}$), the estimator variance explodes as the sample size drops, causing the density manifold $P(x)$ ("position") to dissolve into stochastic Dirac deltas.
2. The Macroscopic Limit ($\sigma_t \to \infty$)**:** We maximize statistical significance ($N_{eff} \to N$), minimizing the Cramér-Rao variance. However, the estimator acts as a low-pass filter, averaging the velocity vector $\mathbf{v}$ to zero. The signal of the regime shift is washed out by the stationarity assumption, effectively "smearing" the market's internal kinetics.

Optimization via Fisher Information Maximization

To rigorously solve for σ_t, I developed a **Frontier Extraction Framework**. Instead of minimizing Mean Squared Error (MSE), which encourages over-smoothing, I seek the "Gabor Optimum"—the bandwidth that maximizes the **Fisher Information Envelope** of the system.

Since the velocity field is the score function $\mathbf{v} = \nabla \ln P$, its magnitude is a proxy for the Fisher Information $I(\theta) = \mathbb{E}[(\nabla \ln P)^2]$. Therefore, the "Constraint Frontier"—the maximum observable velocity at any given density—represents the system's **Thermodynamic Limit**.

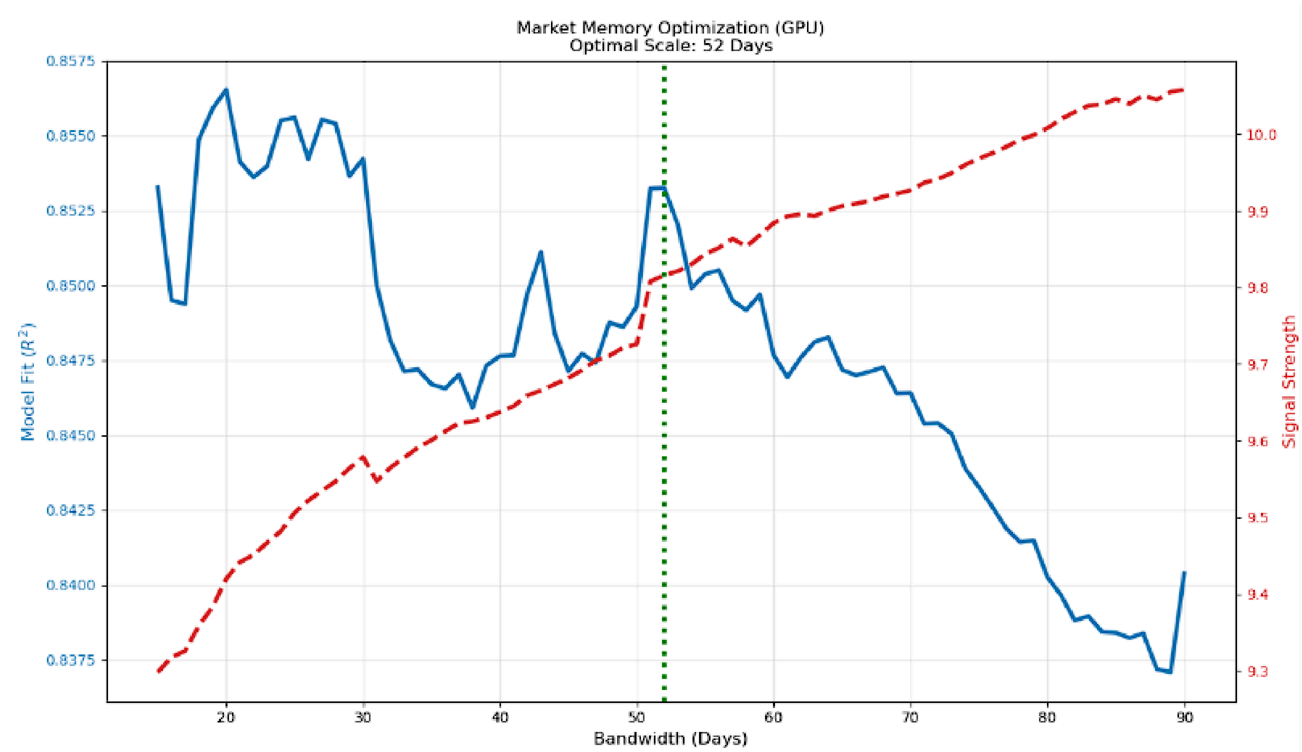
I implemented a GPU-accelerated sweep across the bandwidth spectrum $\sigma_t \in [15, 90]$ days. For each candidate σ_t, I discretized the density field into bins and extracted the 99th percentile of the velocity magnitude $|\mathbf{v}|$ within each bin. The "Signal Strength" was explicitly calculated as the mean of the top 5 velocity peaks along this frontier:

$$S_{signal} = \frac{1}{5} \sum_{i=1}^{5} \max(v)_i$$

This metric isolates the sharpest, most distinct kinematic gradients the model can resolve before they are blurred by smoothing or lost to noise.

The optimization utilized a composite objective function:

$$\text{Score} = \text{Normalized}(R^2) + \text{Normalized}(\text{Signal Strength})$$

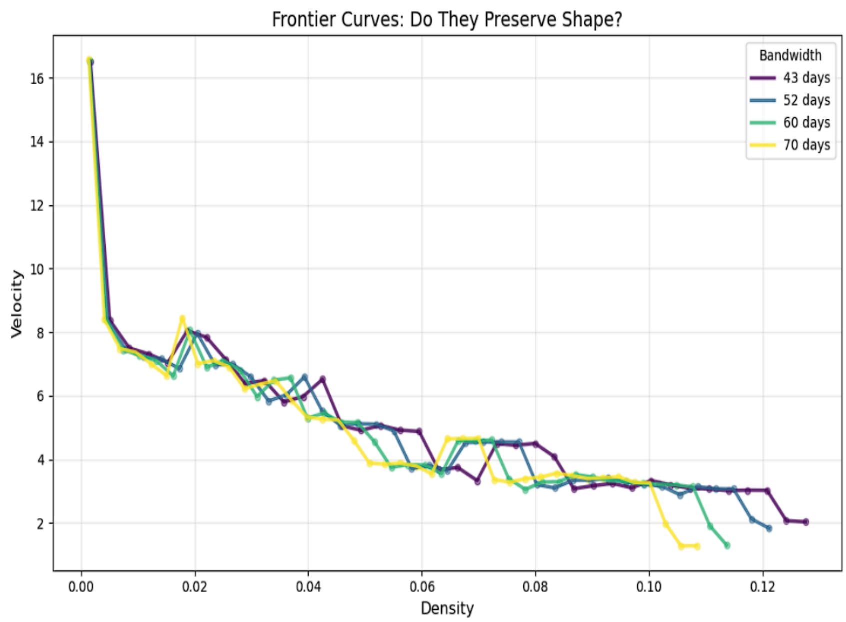


Here, R^2 measures how well the frontier fits a theoretical exponential decay model (validating that the observed structure obeys thermodynamic constraints), while "Signal Strength" measures the raw magnitude of the detected velocity. The algorithm converged convexly on a global optimum of **52 Days**. This value represents the information-theoretic coherence time of the NYC housing market. In other words the average shelf-life of price information; the duration for which a single transaction remains a valid signal for other buyers before fading into 'stale' historical noise. This likely corresponds to contract close latency. Ultimately, this optimization process is akin to a Renormalization Group (RG) flow analysis; by sweeping the temporal bandwidth σ_t, we effectively coarse-grain the microstructure, observing how the Madelung object generalizes across different information scales to separate the relevant macroscopic variables (signal) from irrelevant microscopic degrees of freedom (noise). This is effectively a Kadanoff Block Spin transformation on the market data, averaging out fast fluctuations to see the stable, long-range order

Robustness and Invariance Checks

To ensure that this 52-day scale reflected a genuine property of the market rather than an artifact of the estimator, I performed a "Directional Consistency" test. I generated independent vector fields using bandwidths of 43, 52, 60, and 70 days and calculated the Cosine Similarity of the resulting drift vectors $\mathbf{v}$ at every point in the spatial grid (50×50).

The results demonstrated extraordinary geometric invariance:

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==================================================
DIRECTIONAL AGREEMENT (Cosine Similarity)
==================================================
43 vs 52 days:
  Mean cosine similarity: 0.990
  Fraction agreeing (cos > 0.5): 99.3%

43 vs 60 days:
  Mean cosine similarity: 0.976
  Fraction agreeing (cos > 0.5): 99.1%

43 vs 70 days:
  Mean cosine similarity: 0.958
  Fraction agreeing (cos > 0.5): 98.4%

52 vs 60 days:
  Mean cosine similarity: 0.996
  Fraction agreeing (cos > 0.5): 100.0%

52 vs 70 days:
  Mean cosine similarity: 0.982
  Fraction agreeing (cos > 0.5): 99.5%

60 vs 70 days:
  Mean cosine similarity: 0.994
  Fraction agreeing (cos > 0.5): 99.8%
```

The fact that the vector fields remain directionally locked (mean cosine similarities all exceed 0.95) despite significant changes in bandwidth confirms that the "Madelung Flow" represents a robust, invariant geometric feature of the dataset. The 52-day optimization does not "create" the pattern; it simply brings an existing, invariant geometric structure into maximum focus. This exercise beautifully highlights the parallels between the Cramer-Rao Bound, Heisenberg's Uncertainty, and the bias-variance trade off.

The Heisenberg Limits of Real Estate

Having calibrated the thermodynamic instrument, I turned to the analysis of the phase space itself. The Madelung object provides two scalar fields at every coordinate (p, s) in the market:

Market Certainty (Z): The probability density, representing the consensus on value. High density indicates a "crowded" trade where buyers and sellers agree on price.
Liquidity Velocity ($|\mathbf{v}|$): The magnitude of the drift vector, representing the speed of price discovery or the "momentum" of the asset class.

The Uncertainty Principle of Markets

When visualizing these two variables against each other for any given date, a striking geometric constraint emerges as the economic analogue to Heisenberg's Uncertainty Principle. In quantum mechanics, the precision of position and momentum are inversely related ($\sigma_x \sigma_p \geq \frac{\hbar}{2}$). In the NYC real estate market, we observe a similar fundamental trade-off between Consensus and Velocity.

The Attractor State: Regions of high density (high certainty) effectively "trap" prices. When market participants broadly agree on a value, the "mass" of the consensus creates inertia, preventing rapid price movement.
The Rapid State: Conversely, high velocity is only permissible in regions of low density (low certainty). Rapid price discovery requires a lack of established consensus.

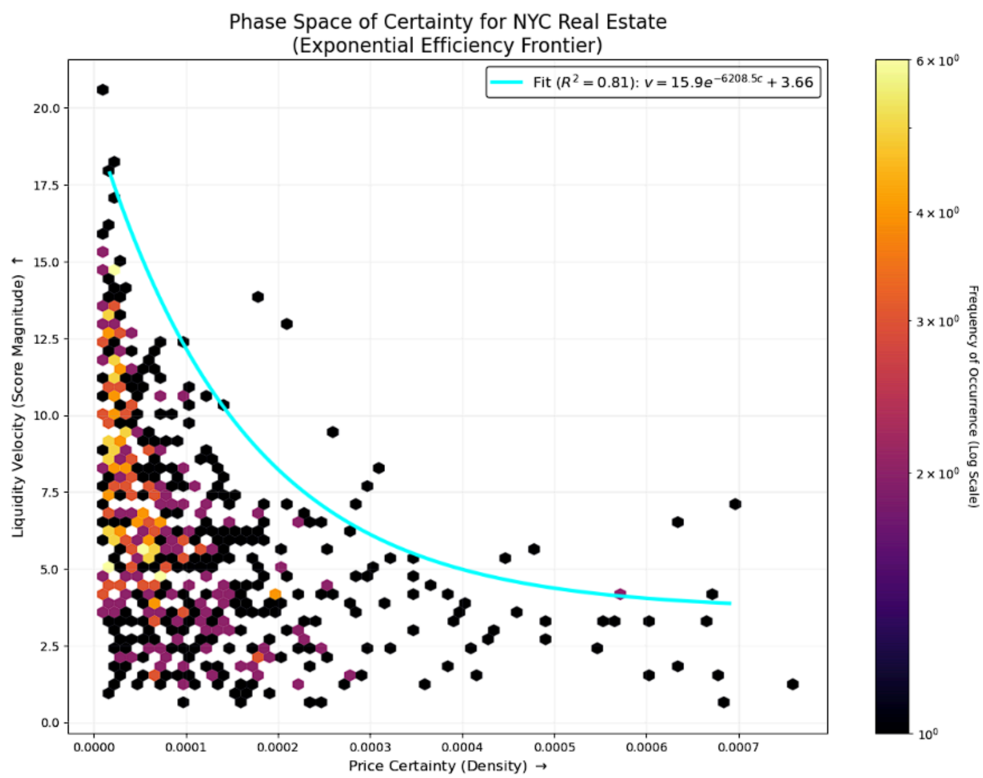
This is not merely a correlation; it is a "forbidden zone" geometry. While outliers exist, the vast majority of transactions adhere to a characteristic efficiency frontier. Beyond this curve, the system exhibits an empirical kinematic limit on its velocity given the weight of the consensus.

The Aggregated Efficiency Frontier

To quantify this limit, I aggregated the phase space data across 40 timestamps (spanning 2015–2024), effectively "superpositioning" the market's history to outline the empirical hard edge of the distribution. I extracted the 99th percentile of velocity within each density bin and fitted an exponential decay model to this frontier:

$$v_{max} = A \cdot e^{-B \cdot Z} + D$$

The global fit yielded a robust coefficient of determination ($R^2 = 0.8125$), suggesting that this trade-off represents a persistent structural regularity within the market's informational manifold, rather than transient stochastic noise.



<u>Global Frontier Metrics:</u>

- **Equation:** $v = 15.87 \cdot e^{-6208.47 \cdot Z} + 3.66$
- **RMSE:** 1.8369

This equation implies a "terminal velocity" for real estate. Even in the most illiquid, chaotic corners of the market (where $Z \to 0$), the speed of price change is capped at approximately **19.5 units**. Conversely, as consensus forms (Z increases), the maximum possible velocity decays exponentially toward a friction floor of 3.66.

Temporal Stability and Parametric Drift

To test the invariance of this, I analyzed the frontier during distinct macroeconomic regimes. While the functional form (exponential decay) remained robust, the specific parameters exhibited a measurable "wobble," reflecting the changing rigidity of the market.

Period A: Pre-Pandemic Stability (2017–2018)

- **Equation:** $v = 16.05 \cdot e^{-23462.00 \cdot Z} + 3.48$
- **Max Speed:** ~19.53
- **Decay:** -23,462

Period B: Pandemic Shock (2020–2021)

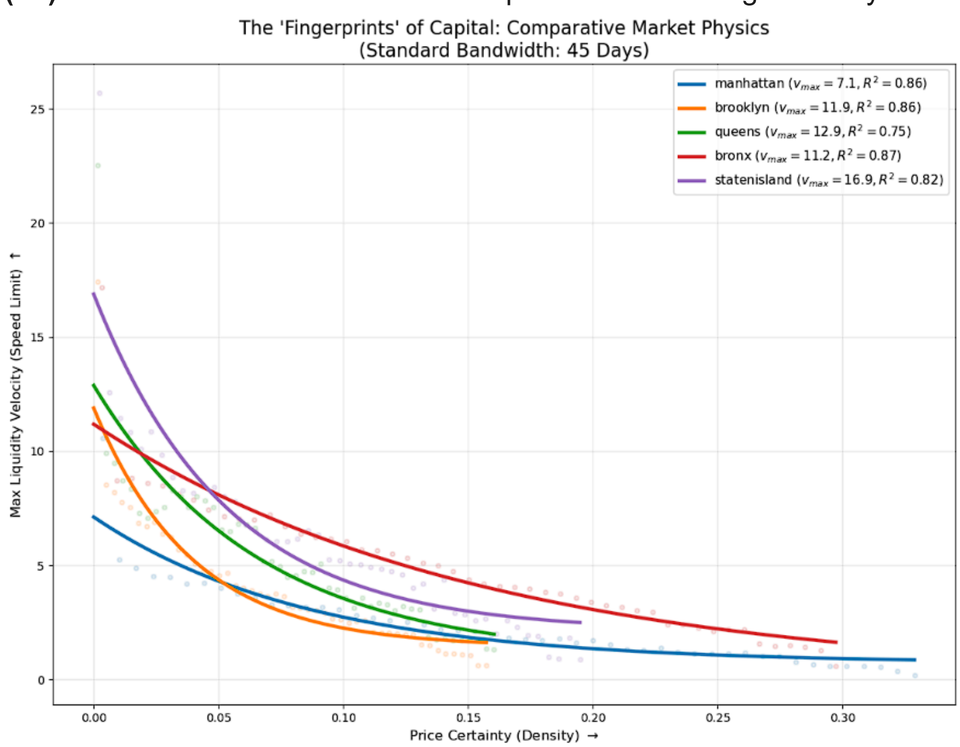
- **Equation:** $v = 11.09 \cdot e^{-8048.87 \cdot Z} + 4.10$
- **Max Speed:** ~15.19
- **Decay:** -8,049

This comparison reveals a fascinating physical transformation. During the stable period (2017–2018), the market was "stiff": the decay coefficient was massive (-23,462), meaning velocity evaporated almost instantly the moment consensus began to form. During the pandemic (2020–2021), the market became "mushier." The maximum speed dropped to 15.19, but the decay was far shallower (-8,049), suggesting that even as consensus formed, higher volatility persisted longer than usual. This parametric drift suggests that while the functional form of the constraint is robust, its scale is context-dependent. Unlike the fixed Planck constant of quantum mechanics, the market's 'minimum unit of action' is a dynamic variable, expanding and contracting with the macroeconomic temperature.

The Fingerprints of Capital: Borough Analysis

While the geometric profile of this trade-off likely remains consistent across different datasets, the specific parameters exhibit geographical variance. By isolating the dataset by borough and fitting the same Heisenberg frontier to each, I extracted the distinct "physical constants" of these sub-markets.

- **Max Speed ($A + D$):** The theoretical speed limit of the market in a vacuum (zero consensus).
- **Fragility (Decay B):** How quickly liquidity freezes as consensus forms. High fragility means the market calcifies rapidly.
- **Friction (D):** The base level of resistance that persists even at high velocity.



BOROUGH	MAX SPEED	FRAGILITY (Decay)	FRICTION (Floor)	R²
manhattan	7.11	11.6	0.71	0.861
brooklyn	11.89	25.4	1.42	0.860
queens	12.88	15.2	0.95	0.752
bronx	11.18	6.5	0.00	0.869
statenisland	16.88	18.9	2.12	0.819

Manhattan: The Heavy Fluid (High Inertia)
Manhattan acts as a heavy fluid with the lowest Max Speed (7.11), indicating extreme inertial mass due to capital density. Even in a vacuum of consensus, the sheer value of assets creates a viscosity that physically prevents rapid acceleration. It appears stable not through efficiency, but through the immense energy required to move prices in either direction.

Brooklyn: The "Snap-Freeze" Phase Transition
Defined by extreme Fragility (25.4), Brooklyn mimics a system prone to instantaneous phase transitions where liquidity collapses the moment consensus forms. This "snap-freeze" behavior reflects a market driven by intense trend following. Unlike the gradual cooling of other boroughs, it locks into rigid price structures immediately upon establishing a baseline.

The Bronx: The Frictionless Anomaly
The Bronx behaves like a superconductor with a unique Friction Floor of 0.00 and the lowest Fragility (6.5). This laminar flow allows momentum to persist longer than anywhere else before settling into a perfect standstill. It stands as the most thermodynamically efficient borough where consensus does not impede flow as drastically as in capital-dense regions.

Staten Island: High-Energy Chaos
As the energetic opposite of Manhattan, Staten Island acts as a "light" fluid capable of explosive price discovery ($v_{max} = 16.88$). However, this comes with high Friction (2.12), reflecting a market of "wares" (unique single-family homes) rather than "shares" (fungible condos). This asset heterogeneity generates irreducible thermal noise, ensuring that even under consensus, the thermodynamic temperature never reaches absolute zero.

Queens: The Hybrid State
Queens presents the lowest goodness-of-fit ($R^2 = 0.752$) because it does not behave as a single thermodynamic material. It likely represents a superposition of two distinct phases of matter, mixing high-density urban centers like Long Island City with vast suburban tracts. This duality muddies the geometric signal, creating a hybrid profile with moderate speed and fragility.

Configurational Entropy:

The Heisenberg analysis in the previous section quantified the kinematics of the boroughs; how fast prices move under varying degrees of consensus. However, to fully understand the material properties of these sub-markets, I also needed to quantify their static structure. To do this, I calculated the Shannon Entropy (S) of the probability density field for each borough. This metric serves as a proxy for Configurational Entropy, measuring the diversity and disorder of the housing stock itself.

Methodology

I utilized the same continuous-time KDE model to generate a probability density surface $\rho(x, y)$ for each borough. To ensure thermodynamic consistency and comparability across regions, I evaluated every borough on the Universal Phase Space grid defined in the global analysis.

The continuous density field was discretized into a probability mass function $P(x_i, y_j)$ such that $\sum P = 1$. The structural entropy was then calculated using the standard Shannon formulation:

$$S = -\sum_{i,j} P(x_i, y_j) \ln P(x_i, y_j)$$

A high entropy value indicates a "disordered" market occupying a vast region of the phase space (e.g., a mix of micro-studios, luxury penthouses, and brownstones). A low entropy value indicates a "crystalline" market, where transactions are tightly clustered around a specific asset type (e.g., standardized single-family homes). The analysis revealed a stark hierarchy of disorder across the city. Manhattan exists in a state of extreme entropic disarray, while Staten Island represents a highly ordered lattice.

Configurational Entropy by Borough:

	Borough	Entropy (S)	Peak Density
0	manhattan	5.656414	0.177895
1	brooklyn	4.364269	0.631871
2	bronx	4.002923	1.146864
3	queens	3.760683	1.016924
4	statenisland	3.544893	1.238102

This structural data provides the critical context needed to interpret the Heisenberg results. It resolves the apparent paradox of why Manhattan (High Entropy) moves slowly, while Staten Island (Low Entropy) moves fast.

Manhattan (The Viscous Glass): Manhattan combines high structural disorder ($S = 5.66$) with high inertia. Physically, this describes a Viscous Glass. Like a glass, it lacks a defined crystalline structure (diverse assets), but its viscosity prevents it from flowing freely. It is frozen by its own complexity.

Staten Island (The Hot Crystal): Staten Island combines high structural order ($S = 3.54$) with high kinetic velocity. Physically, this describes a Hot Crystal. The lattice is uniform (standardized housing stock), but the individual particles are vibrating violently (high price volatility).

The Bronx (The Superfluid): The Bronx exhibits the ordered structure of a crystal ($S = 4.00$) but the flow of a superfluid (0.00 friction). This suggests a Laminar Flow regime where standardization allows for frictionless price discovery.

Brooklyn (The Supercooled Liquid): Brooklyn sits in the middle of the entropy spectrum ($S = 4.36$) but has the highest Fragility (25.4). It behaves like a Supercooled Liquid, a state that is technically fluid but prone to "snap-freezing" into a rigid solid structure the moment a trend appears. The moderate asset diversity allows for both fluid price discovery and sudden rigid consensus.

Queens (The Polycrystalline Hybrid): Queens remains the anomaly with the lowest goodness-of-fit ($R^2 = 0.75$). Its low entropy ($S = 3.76$) suggests order, but its mechanics are inconsistent. It likely behaves as a Polycrystalline Solid, a material composed of many small, distinct ordered domains (e.g., rigid Long Island City condos vs. fluid Jamaica estates) that do not align along a single thermodynamic grain, scattering the geometric signal.

Visualization of the Probability Field

With the mathematical framework established and calibrated, the next step of the pipeline involves rendering the Madelung Object $M(x)$ into a human-interpretable format. I developed a suite of visualization scripts using Plotly (for interactive HTML exploration) and Matplotlib (for high-resolution GIFs) to map the evolution of the probability fluid over the decade.

The Flowing Market (4D Surface Animation)

The primary visualization renders the scalar field $\rho(p, s, t)$ as a dynamic 3D surface, effectively visualizing the real-time flows of liquidity through the market.

Grid Construction: I defined a 50×50 mesh grid spanning the log-space of Building Area (X) and Price Per Square Foot (Y).

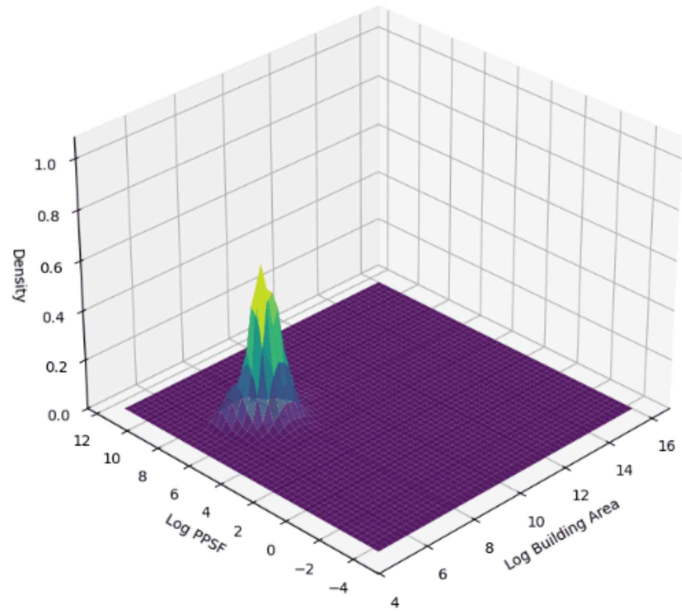
Time-Stepping: The script iterates through the dataset in 14-day increments. For each time step, the 'ContinuousTimeKDE' object (tuned to $\sigma_t = 52$ days) calculates the density Z at every grid point using the weighted Gaussian summation.

Rendering: The Z-axis represents Market Consensus (Transaction Density).

Peaks (High Z): Correspond to "Attractor" prices, regions of high liquidity where the volume of transactions is dense. These are the "popular" trades (e.g., standard condos in mid-tier neighborhoods).

Valleys (Low Z): Represent illiquid or forbidden zones where capital rarely flows.

Motion: As the animation plays, we see the probability mass "slosh" and migrate. This visualization directly confirms the "Parametric Drift", showing the market physically expanding and contracting; showing the flows of liquidity throughout the market over time.

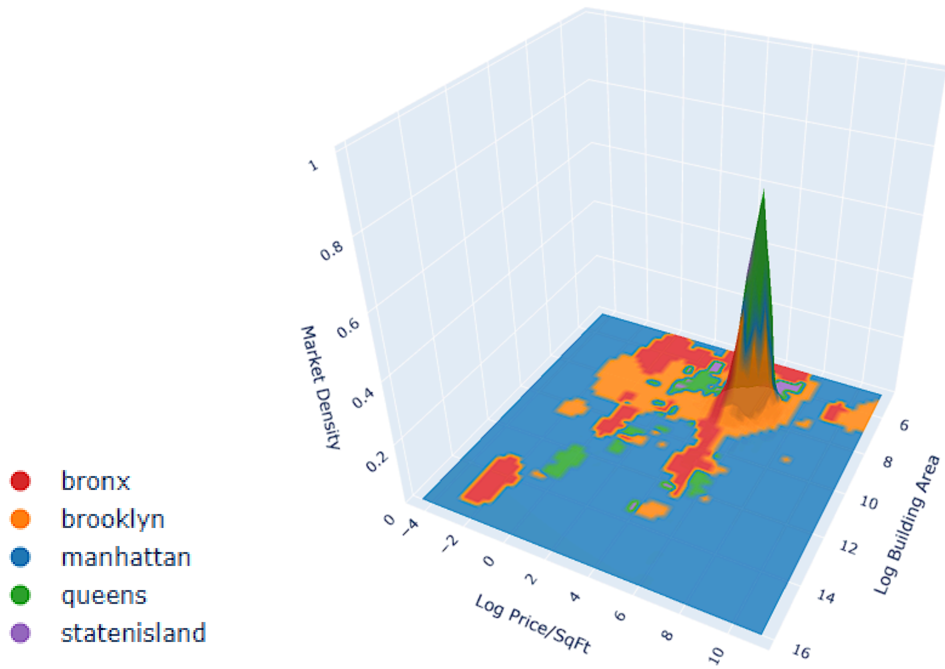


Borough Territories (Multi-Class Topology)

To visualize the structural heterogeneity, I implemented a Multi-Class KDE engine. This script creates a topological map of "Borough Dominance," showing which geographic sub-market controls which region of the price/size phase space.

- The Global Manifold: First, the script calculates a global density surface Z_{total} using all transactions, regardless of location. This defines the overall shape of the liquidity mountain range.
- The Winner-Takes-All Color Map: For every point on that grid, the script calculates the localized density of each individual borough. It then applies a np.argmax function to determine which borough possesses the highest probability density at that specific price/size coordinate.
- Surface Mapping: The visual output plots the global Z surface but colors the surface of the manifold based on the dominant borough.

The result is a 3D territorial map. We can see how all boroughs share the and swap pieces of the peak, with Manhattan dominating the areas outside of it with low transaction occurrence, confirming the high entropy measured in that borough earlier.



The Visualization of Forces

While the probability density ρ tells us where the market is, it is the hydrodynamic derivative fields that tell us what the market is doing. In the Madelung picture, the evolution of the market is governed by a continuity equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0$$

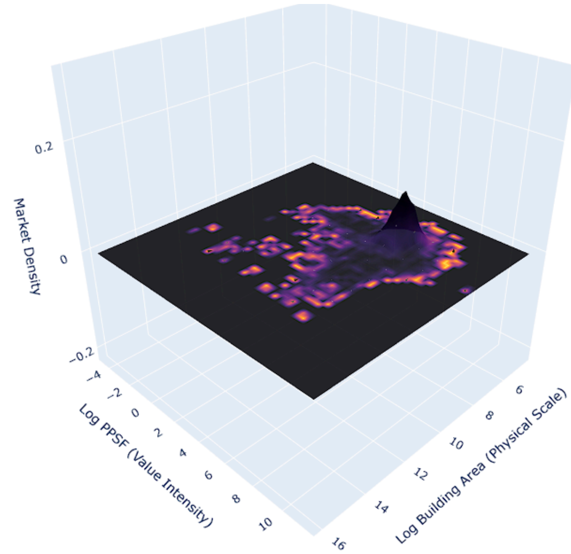
Here, the velocity field $\mathbf{v} = \nabla \log \rho$ represents the "drift" of the probability mass. I developed a series of visualizations to isolate and interpret the specific components of this drift.

The Metric Instability Field ($|\mathbf{v}|^2$)

The first derived quantity is the squared magnitude of the velocity vector, $|\mathbf{v}|^2$. In information geometry, this is proportional to the trace of the Spatial Fisher Information Metric. Think of Fisher Information as reaching into a bag of mixed marbles: if the mix is 50/50, pulling out a red marble tells you very little (low information, flat gradient); but if the bag is 99% blue, finding a red marble is a massive signal (high information, steep gradient). Unlike traditional FIM, which measures how sensitive the data is to hidden *parameters* (the mix ratio), the Spatial FIM measures how sensitive the probability is to location, specifically, how rapidly the market regime shifts as you move through price/size space. Because our velocity vector is defined as the gradient of the log-density ($\mathbf{v} = \nabla \log \rho$), squaring this vector ($|\mathbf{v}|^2$) naturally recovers the Fisher Information metric, confirming that 'speed' in this model is literally the speed of information update.

Code Implementation: I calculated the analytic derivatives U and V from the KDE and computed their squared sum at every grid point. To ensure visual clarity, I clipped the values at the 98th percentile; without this, a single extreme outlier (a massive price jump) would "wash out" the color scale, hiding the structural details of the rest of the market.

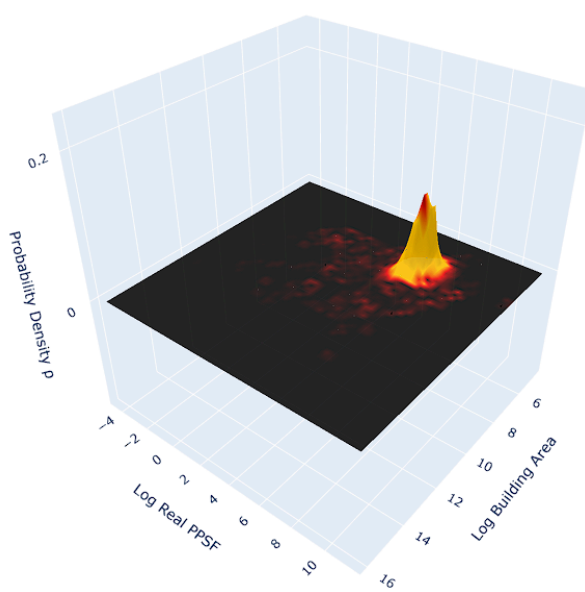
<u>Physical Interpretation:</u> I termed this quantity "Metric Instability." It measures the local speed of the log-likelihood surface. Regions with high instability are zones of Rapid Information Update. Mathematically, these are areas where small perturbations in position result in massive changes in probability; effectively, the "event horizons" of price discovery.



Information Kinetic Energy Density ($\rho|\mathbf{v}|^2$)

While Metric Instability tells us where the potential for movement is high, it does not tell us where the economic work is actually being done. For that, we need the Information Kinetic Energy Density, calculated by weighting the instability by the probability density or "mass" of the market. This quantity has a direct antecedent in quantum chemistry: von Weizsäcker's (1935) kinetic energy functional for electron densities, $T_W \propto \int|\nabla\rho|^2/\rho$, which was later recognized as mathematically equivalent to the Fisher Information (Nalewajski, 2003). Probability plays a role analogous to energy as a conserved accounting quantity governing admissible transformations. This quantity is not interpreted as physical energy, but as a geometric diagnostic of information flow intensity.

$$K(x) = \rho(x)|\mathbf{v}(x)|^2$$

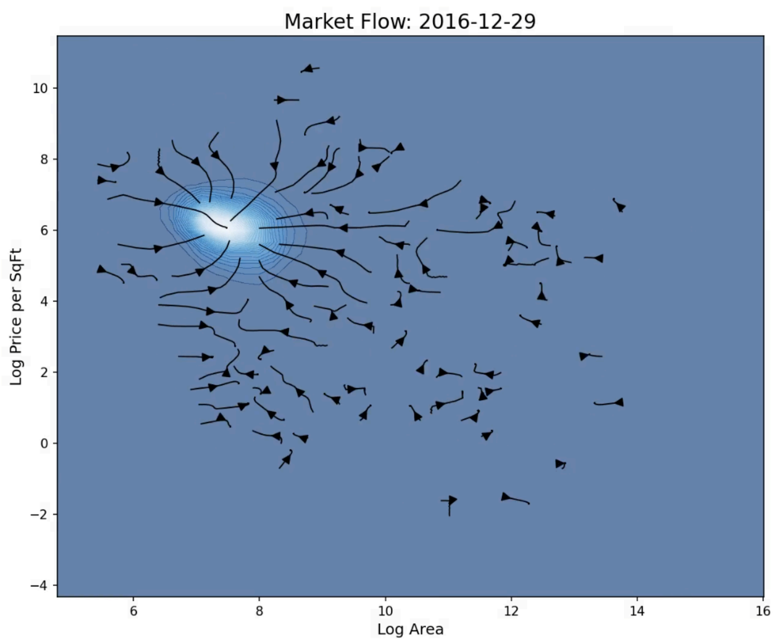


The visualization revealed a fascinating structural phenomenon; a ring of fire.
The Core (Pale Peak): The absolute center of the density mountain (high ρ) actually has low kinetic energy. This is the "dead center" of consensus; transactions are frequent, but they occur at a stable, agreed-upon price.

The Periphery (Dark): The far edges have high velocity (instability) but zero density. No work is done here because there are no transactions to carry the energy.

The Shoulder (Bright Ring): The Kinetic Energy peaks in a distinct ring surrounding the core. This is the Active Transport Zone. It is the boundary where substantial transaction volume meets substantial price disagreement. This visualization confirms that the "thermodynamic work" of the market, the actual effort of price discovery, occurs not at the stable center, but at the negotiating fringe.

The Gradient Flow of Information



To visualize the directionality of these forces, I mapped the gradient field $\nabla \log \rho$ using 2D streamlines.

Code Implementation: Using matplotlib.streamplot, I integrated the velocity vectors (U, V) to trace the "currents" of the probability fluid.

Geometric Interpretation: They show the natural downhill paths of the market, pointing in the direction of least suprise. In a perfectly efficient market, these lines would converge radially on the attractor. In the NYC data, however, we see distinct vortices and "shear zones" where the flow of capital is turbulent, particularly in the transition zones between boroughs (e.g., the boundary between high-end Brooklyn and low-end Manhattan).

The Quantum Potential as Information Curvature

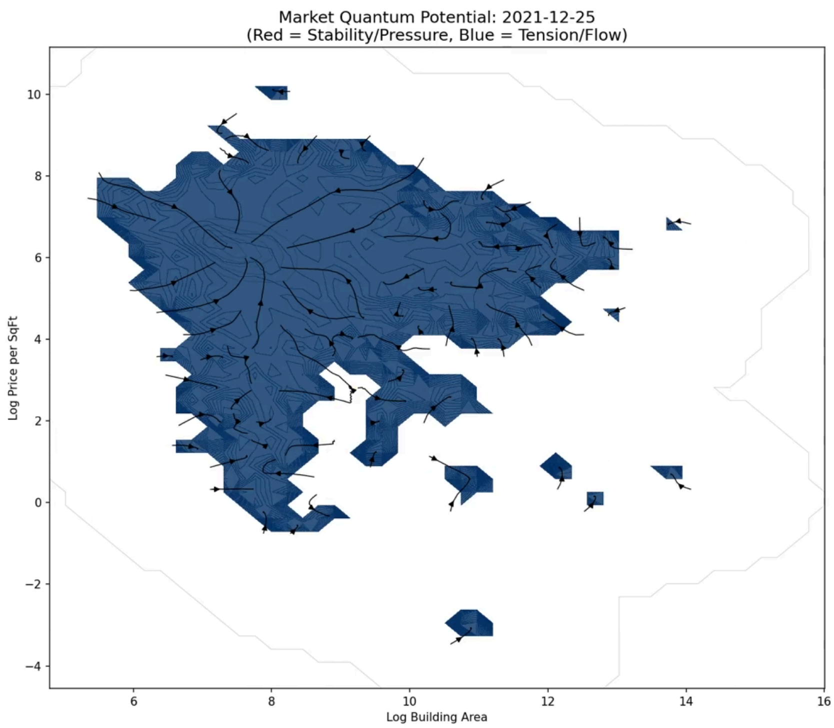
I visualized the Quantum Potential (Q), revealing its explicit identity as a geometric object. In the Madelung formulation, Q is not a mysterious force, but an algebraic functional of the score function or velocity vector:

$$4\frac{\nabla^2\sqrt{\rho}}{\sqrt{\rho}} = |\nabla \ln \rho|^2 + 2\nabla^2 \ln \rho$$

This decomposition makes the bridge to Information Geometry mathematically transparent by separating the potential into two distinct statistical descriptors:

- The Fisher Information Component ($|\mathbf{v}|^2$): The term $|\nabla \ln \rho|^2$ represents the squared norm of the score function. In information theory, the expectation of this term is the Fisher Information, which measures the sensitivity of the probability density to changes in its parameters. Locally, it defines the metric curvature of the state space.
- The Diffusion/Divergence Component ($2\nabla^2 \ln \rho$): The term $2\nabla^2 \ln \rho$ (or $2\nabla \cdot \mathbf{v}$) represents the divergence of the score. This is the Laplacian of the log-density, which characterizes the local concentration or dissipation of information.

Thus, the Fisher Information is effectively the unscaled, divergence-free component of the Quantum Potential. Think of Q as the 'cost of certainty'; just as you cannot squeeze a balloon into a single point without it pushing back, Q is the repulsive force that engages whenever the market tries to form a perfect consensus, preventing the probability cloud from collapsing into a mathematical singularity.



This analysis reveals the New York City housing market is always in a flow state. It is important to note here the quantum potential used in this analysis is a statistical curvature term, not a physical force. It represents the local deviation from a classical equilibrium, manifesting as an informational pressure that guides the trajectory of value through the phase space.

Note on Scaling: While the geometric identity of Q is precise, its absolute magnitude in a physical or socio-economic system depends on a coupling constant; an effective Planck parameter ($\hbar_{eff}$). In his seminal 1966 work, Edward Nelson demonstrated that the Schrödinger equation could be derived from classical stochastic mechanics by treating the diffusion coefficient (ν) as the fundamental scaling factor, where the relationship is typically defined as $\hbar = 2m\nu$.

In the context of this study, this implies that the "quantum" behavior of the Madelung object is not a fixed universal, but is instead modulated by a local, time-dependent scalar representing the system's "volatility temperature." While Nelson's derivation opens a clear path for future analysis, specifically in using empirical market volatility to calibrate the strength of the informational potential, such work is beyond the scope of this initial framework. Here, Q is presented to establish the formal bridge between Bohmian mechanics and Information Geometry, rather than as a calibrated empirical signal.

Theoretical Extension: Lie Derivatives and Noether Symmetries

The construction of a differentiable manifold theoretically permits a rigorous search for hidden conservation laws using Lie Derivatives ($\mathcal{L}_{\mathbf{v}}$). By computing how the Fisher Metric g distorts as it flows along the velocity field, one can hunt for Noether Symmetries; mathematical proofs that a quantity (like "market energy") is conserved over time. Specifically, a vanishing Lie Derivative ($\mathcal{L}_{\mathbf{v}} g = 0$) would identify a "Killing Vector Field," implying that the market's information geometry is invariant under evolution. Conversely, a non-zero result quantifies the system's dissipation or entropy production. While preliminary experiments were conducted to test these geometric symmetries, they remain incomplete and were excluded from this analysis for the sake of brevity. However, this line of inquiry is of an extreme significance; these tests can reveal the fundamental rules a system follows, or has followed. The following article explains the concept quite well in an accessible way:
https://www.quantamagazine.org/what-are-lie-groups-20251203/

Theoretical Extension: Geometric Identification & Holographic Elasticity via the Tangent Space

A fundamental limitation in classical econometrics is the "Identification Problem": the inability to distinguish a shift in the Demand curve from a shift in the Supply curve using only equilibrium transaction data (P, Q). Traditionally, this requires "natural experiments" or instrumental variables to artificially hold one curve constant while the other moves.

The Madelung Object offers a radical alternative to bypass this identification problem: Geometric Identification. By elevating the analysis from discrete points to a continuous probability manifold, we bypass the need for external controls. Instead, we extract elasticities directly from the Tangent Space of the density field.

The Iso-Probability Contour as the Indifference Curve

Consider the density field $P(A, \pi, t)$, where A is Log-Area and π is Log-Price. The "Iso-Probability Contours" (lines where $P = \text{constant}$) represent the market's revealed indifference curves. They define the set of Price-Size combinations that the market considers equally valid (equally liquid) at time t.

By the Implicit Function Theorem, the slope of this contour at any coordinate represents the Local Effective Elasticity (ϵ_{eff}). It quantifies the instantaneous trade-off between size and price that preserves the transaction probability:

$$\left.\frac{d\pi}{dA}\right|_{P=const} = -\frac{\partial P/\partial A}{\partial P/\partial \pi} = -\frac{\mathbf{v}_{osm}^{(A)}}{\mathbf{v}_{osm}^{(\pi)}}$$

Here, $\mathbf{v}_{osm}$ is the Osmotic Velocity vector (the structural gradient).

- **If the ratio is High (Vertical Slope):** The market demands a massive price increase for a small size increase (Inelastic Supply/High Scarcity).
- **If the ratio is Low (Horizontal Slope):** The market allows size to increase with minimal price change (Elastic Supply/High Inventory).

Vector Decomposition of Market Forces

Unlike a static regression which gives a single elasticity coefficient for the whole market (e.g., "Demand Elasticity is -0.5"), the Madelung Object yields a Tensor Field of Elasticities. We can calculate the elasticity vector at *every* point in the grid.

Furthermore, by decomposing the total flow $\mathbf{J}$ into its temporal components, we can isolate Dynamic Elasticity (how prices react to flow) from Structural Elasticity (how prices relate to size).

$$\epsilon_{dynamic} = \frac{\partial \pi}{\partial t} \Big/ \frac{\partial A}{\partial t}$$

This formulation allows us to observe the "breathing" of supply and demand constraints in real-time. We do not need a control group to see if a price hike caused a drop in volume; we simply observe the divergence of the probability flux ($\nabla \cdot \mathbf{J}$). If the flux diverges (negative source term) in response to a price move, the demand destruction is geometrically observable as a "dissipation of probability."

This suggests that the Madelung Object acts as a Holographic Observer: it encodes the full structural information of the latent Supply and Demand surfaces into the curvature of the observable transaction manifold, rendering the "Identification Problem" an artifact of low-dimensional (discrete) observation methods. This will be fully demonstrated in upcoming work.

The Bataillean Decomposition

The pan-ultimate experiment of this study bridges the gap between thermodynamic physics and heterodox economics. I would consider it the crown jewel of what I have so far. It is inspired by the work of French philosopher Georges Bataille, whose 1949 treatise *The Accursed Share* proposed a “solar economy” rooted in the physical reality that the sun floods the earth with a constant, unreciprocated superabundance of energy, implying that the fundamental drive of any system, biological or economic, is not the accumulation of scarce resources, but the necessary dissipation of inevitable excess energy. He categorized this expenditure into two forms:

Growth (Structure): Energy used to expand the system or maintain its “form”.

Waste (Luxury): The "accursed share" of energy that cannot be absorbed by growth and must be catastrophically expended (through luxury, war, or non-productive consumption) to prevent the system from exploding. Bataille's paradigmatic example was the potlatch; the ceremonial destruction or gifting of wealth practiced by Indigenous peoples of the Pacific Northwest, where surplus was ritually burned or given away precisely because it could not be productively absorbed.

While dismissed by mainstream economists of his day, Bataille's intuition aligns remarkably with modern Stochastic Thermodynamics. In the Fokker-Planck framework, the total entropy production of a system can be mathematically decomposed into two distinct components: the "Housekeeping Heat" (energy dissipated to maintain non-equilibrium steady states) and the "Excess Heat" (energy dissipated during transitions between states).

Mathematical Formulation

We begin with the continuity equation, which links the evolution of the density $\dot{P}$ to the divergence of a probability current J:

$$\frac{\partial P}{\partial t} + \nabla \cdot \mathbf{J} = 0$$

Unlike standard decompositions which separate flow into rotational (eddies) and irrotational parts, my model imposes a Minimum Kinetic Energy Constraint. We solve for the unique, curl-free current $\mathbf{J}$ that strictly satisfies the observed time evolution. This effectively ignores "invisible" circular trading loops that do not alter the price distribution, isolating only the flows that result in structural change.

To do this, I solve the Poisson equation for a scalar potential field ϕ:

$$\nabla^2 \phi = \frac{\partial P}{\partial t}$$

This is solved efficiently over the 2D market grid using the Fast Fourier Transform (FFT). In the frequency domain, the differential operator becomes algebraic division, allowing us to invert the Laplacian and recover the potential ϕ. The total current is then the gradient of this potential: $\mathbf{J}_{Vdisp} = -\nabla \phi$.

We then separate the market's total velocity field into two orthogonal thermodynamic components:

1. The Osmotic Velocity ($\mathbf{v}_{osm}$): The Cost of Structure

$$\mathbf{v}_{osm} = D \nabla \ln P$$

This velocity represents the "entropic pressure" required to maintain the shape of the probability cloud against the diffusing force of noise. It exists even when the market is static. In Bataille's framework, this is The Cost of Structure—the continuous energy expenditure required to uphold the hierarchy of prices (e.g., maintaining the premium of Manhattan over the outer boroughs). It is the energy of "staying the same."

2. The Displacement Velocity ($\mathbf{v}_{disp}$): The Accursed Share

$$\mathbf{v}_{disp} = \frac{\mathbf{J}}{P}$$

This velocity exists *only* when the probability distribution is changing ($\dot{P} \neq 0$). It represents the physical transport of liquidity from one regime to another. In Bataille's framework, this is The Accursed Share. It is the surplus energy that disrupts the status quo, forcing the market to move. A crash or a boom is not an error; it is a Potlatch—a necessary, intense expenditure of energy to dissipate the information that the current structure could not absorb.

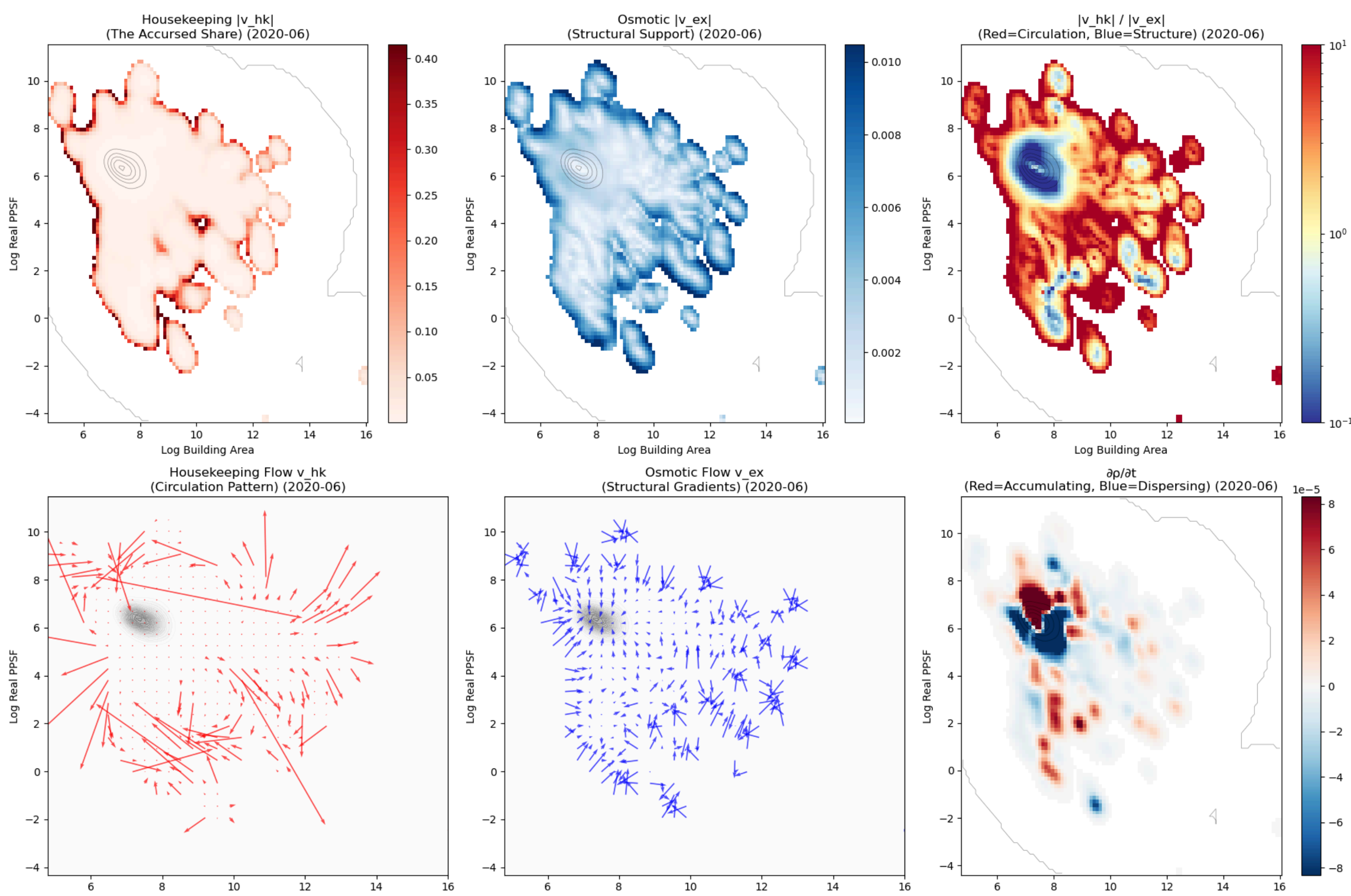
The Bataille Ratio and the Potlatch Condition:

By integrating the thermodynamic cost of these two velocities, we derive a dimensionless indicator of market fragility, the Bataille Ratio (β):

$$\beta = \frac{\text{Entropy of Displacement (Accursed Share)}}{\text{Entropy of Structure (Maintenance)}}$$

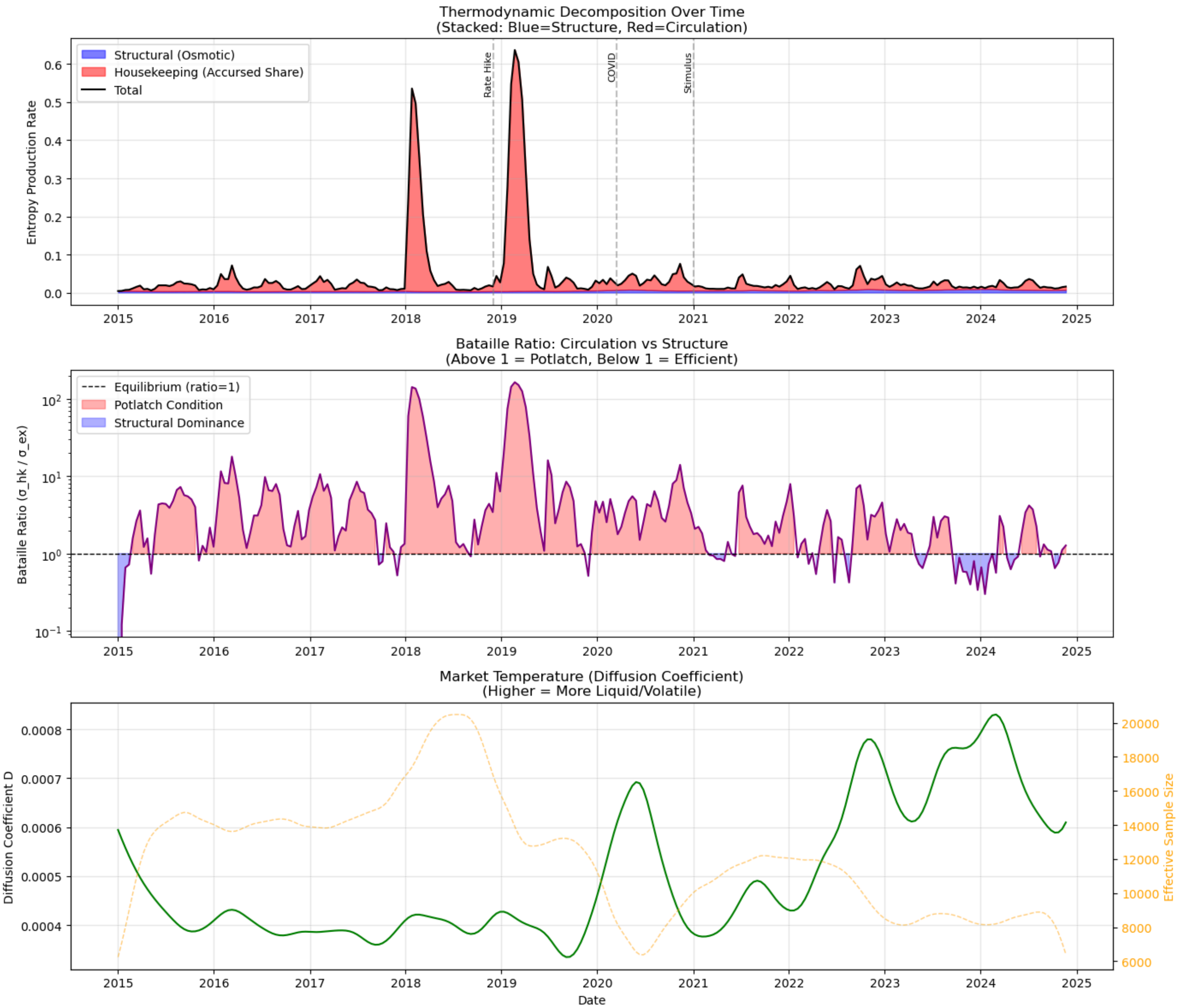
- $\beta < 1$ <u>(Structural Dominance):</u> The market dissipates more energy maintaining its current prices than changing them. The regime is stable.
- $\beta > 1$ <u>(The Potlatch Condition):</u> The cost of moving the market exceeds the cost of maintaining it. The "Accursed Share" has overwhelmed the structure. This signals a phase transition where the market is actively destroying its old geometry to build a new one.

Note on terminology:
The Osmotic Velocity is also often referred to as “excess” heat or velocity in thermodynamics literature. While Bataille refers to non-productive expenditure as "excess," this term more effectively maps to the osmotic velocity following stochastic mechanics convention. The Bataillean "accursed share" corresponds to our displacement velocity component (also called housekeeping), which represents circulation that cannot be derived from a scalar potential - the truly dissipative, non-conservative flow.



The "Potlatch" Condition

The decomposition revealed that for 81.1% of the observed period (2015–2021), the market operated in a "Potlatch Condition".



This implies that the vast majority of the 'work' done by the NYC real estate market consists of excess expenditure over structural maintenance. This fits with the assumption of a liquid market, the system spends significantly more energy churning transaction volume (maintaining the flow) than it does maintaining the rigidity of the price hierarchy (preserving the form against diffusion). We are observing a system optimized to be a luxury engine to dissipate global surplus capital (Potlatch), rather than a market designed to efficiently house people. The market could be described as homeostatic, constantly burning energy to maintain capital flows. The osmotic velocity may be reflective of zoning constraints and geography (the things you cannot do), as well as consumer taste, preferences/utility (the things you don't want to do) the limiting properties and factors of the system seeking to hold it still, while the displacement velocity prevails as the flow of money and people through the system. The analysis shows that the thermodynamic force of capital dominates roughly 4 times as often compared to the force of regulatory barriers and constraints. The danger isn't the market burning energy, it is when it stops. When the ratio drops and structure takes over it means the flows of liquidity in the market have dried up, indicative of a crash or downward correction. In Bataillean terms, the system collapses not from 'waste,' but from a cessation of the necessary expenditure required to sustain its life.

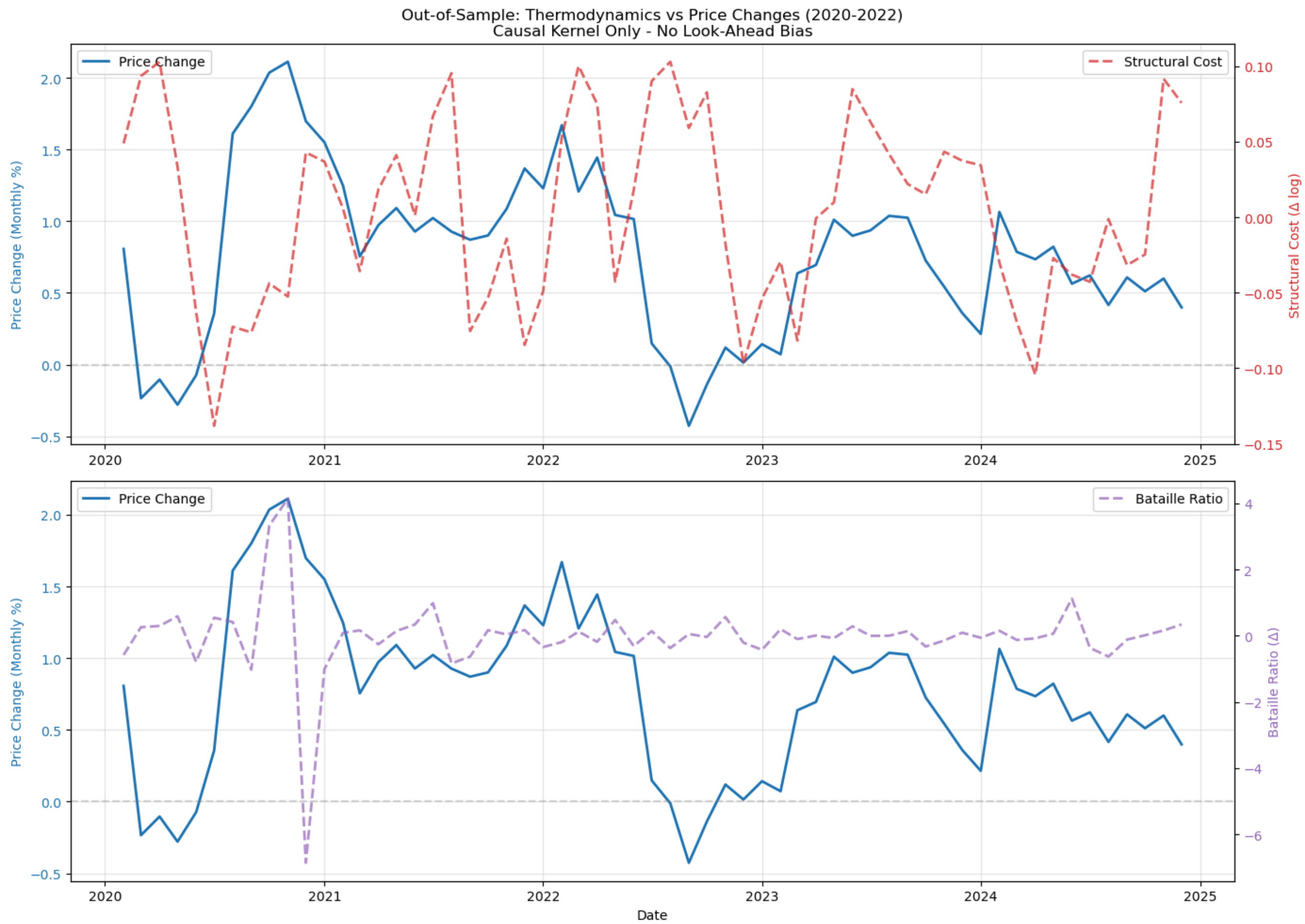
Empirical Validation: "Structural Cost" as a Leading Indicator

To test if this thermodynamic abstraction had predictive power in the real world, I performed a Granger Causality Test against the Case-Shiller NYC Home Price Index. The Case-Shiller index tracks aggregate home price appreciation using a repeat-sales methodology, measuring how the same properties change in value over time. While the underlying transaction data overlaps with the Madelung object, the two constructions measure fundamentally different quantities: Case-Shiller captures price levels, while Structural Cost captures the geometric stress of maintaining the market's distributional structure. The hypothesis was simple: Does thermodynamic stabilizing ("Structural Cost") precede price corrections? To ensure strict statistical integrity, this test utilized a Causal Kernel, reconstructing the density manifold at each time step using only historical data available up to that moment, eliminating any look-ahead bias.

Test Results:

Granger Causality: The "Structural Cost" metric statistically predicts changes in the Case-Shiller Index with a 1-month lag at high significance ($p = 0.0019$) and this significance persists with the 2 month lag as well ($p = 0.0127$).

Predictive Regression (OLS): The Ordinary Least Squares (OLS) model yielded an adjusted R^2 of 0.705. The coefficient for Structural Cost was negative and significant ($\beta = -0.0223$, $p = 0.002$).



This confirms a robust informational constraint in the underlying data: thermodynamic friction is a precursor to price correction, distinct from simple momentum. When the system generates high structural costs, representing the entropic cost of maintaining structural rigidity, it enters a state of unsustainable stress. The market resolves this instability through a 'cooling event,' shedding value to lower the energy required to sustain its form. Thus, Bataille's concept is transformed from a philosophical metaphor into a rigorous, predictive quantity; the structural overheating of the market is a tangible signal of impending regime change.

A supplementary regression excluding price momentum isolated the pure thermodynamic signal, yielding an R^2 *of 0.22 (*$p < 0.01$*). Notably, the predictive power shifted to a 2-month lag (*$\beta_{t-2} = -0.035$*), suggesting a characteristic delay between peak structural stress and the subsequent price cooling.*

```
==============================================================================
                            OLS Regression Results
==============================================================================
Dep. Variable:           Price_Change   R-squared:                       0.720
Model:                            OLS   Adj. R-squared:                  0.705
Method:                 Least Squares   F-statistic:                     46.31
Date:                Fri, 23 Jan 2026   Prob (F-statistic):           5.96e-15
Time:                        23:12:39   Log-Likelihood:                 253.15
No. Observations:                  58   AIC:                            -498.3
Df Residuals:                      54   BIC:                            -490.1
Df Model:                           3
Covariance Type:            nonrobust
==========================================================================================
                          coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------
const                   0.0017      0.001      2.396      0.020       0.000       0.003
Structural_Cost_Lag1   -0.0223      0.007     -3.194      0.002      -0.036      -0.008
Bataille_Lag1        4.487e-06      0.000      0.013      0.990      -0.001       0.001
Price_Lag1              0.7796      0.073     10.639      0.000       0.633       0.926
==============================================================================
Omnibus:                        4.446   Durbin-Watson:                   1.918
Prob(Omnibus):                  0.108   Jarque-Bera (JB):                5.290
Skew:                           0.035   Prob(JB):                       0.0710
Kurtosis:                       4.478   Cond. No.                         212.
==============================================================================
```

Technical Note: Thermodynamic Nomenclature

A semantic distinction exists between standard Non-Equilibrium Thermodynamics (NET) and the "Bataillean" interpretation employed here. While strict NET frameworks (e.g., Oono & Paniconi) define "Housekeeping Heat"/"Displacement Velocity" as the dissipation required to maintain steady-state currents, I mapped this term to Bataille's "Accursed Share"; the energy dissipated purely to sustain market circulation without altering its configuration. Conversely, I associate the "Osmotic" or "Structural Cost" term with the maintenance of structure, quantifying the energy expended to uphold probability gradients, the score, against the eroding force of diffusion. This alignment preserves the mathematical rigor of the flux decomposition while correctly operationalizing Bataille's sociological distinction between the energy required to maintain form and the surplus that must be expended to prevent systemic collapse.

<u>On the Isomorphism to Diffusion Models:</u>

This framework aligns closely with score-based diffusion models in that it estimates the score function $\nabla \log p(x,t)$ governing the evolution of probability mass. Unlike neural diffusion models, this score is inferred nonparametrically via a differentiable kernel density estimator, enabling explicit recovery of drift fields, diffusion tensors, and probability currents. This places the method within the class of information-geometric diffusion learning approaches, where latent dynamics are inferred directly from evolving distributions rather than parameterized generative models. There is indeed a potential to accelerate this framework and eliminate the curse of dimensionality that somewhat shackles this approach in its current form through the development of a "Madelung Flow Denoising" model.

<u>Conclusion:</u>

This study modeled the New York City real estate market as a continuous probability fluid to reveal three distinct structural regularities observed within the inferred manifold across distinct scales. Through the construction of the Madelung Object, I identified a characteristic coherence scale of 52 days required for signal stability, an informational constraint analogous to the Heisenberg limit governing the trade-off between price certainty and liquidity, and a dissipative regime (the 'Potlatch Condition') where the market maintains structural dominance through the expenditure of excess informational flux for the vast majority of the decade. These findings move beyond standard econometric correlations to offer a rigorous physical description of the geometric and dynamic constraints of the market.

The dominance of displacement velocity in the flux decomposition fundamentally reframes the economic problem from one of scarcity to one of dissipation. Consistent with Georges Bataille's concept of a solar economy, the data suggests the market functions not as an efficient allocator of scarce resources but as a dissipative system driven by the necessary expenditure of excess energy. The validation of Structural Costs as a leading indicator ($p = 0.002$) confirms that price corrections are thermodynamic cooling events which are structurally necessary to restore flow when the structural tension of the system exceeds its capacity to dissipate.

While this initial model establishes a robust framework, it likely suffers from omitted variable bias as it treats the housing market as a closed thermodynamic system without explicit coupling to external macroeconomic rate variables or policy shocks. However, this limitation highlights the flexibility of the model as it is designed to be modified and extended. This specific implementation represents the culmination of over a year of experimentation by synthesizing insights derived from separate and parallel investigations into the thermodynamics of the stock market and inflationary dynamics into a single cohesive theory of information geometry.

The utility of this hydrodynamic approach extends far beyond real estate. The mathematical machinery developed here to solve for probability currents and potentials is broadly applicable across the sciences and offers new tools for modeling phenomena ranging from the Navier-Stokes equations in fluid dynamics and turbulence to complex interactions in ecology and medical econometrics. By monitoring the invisible gradients of the probability field, this framework provides a viable instrument for Economic Seismology that is capable of detecting the structural heat of a crisis long before it manifests as a crash in price.

Acknowledgements:
I would like to acknowledge Dr. Estelle Epstein at the Rutgers Newark Physics Department for her curiosity and support throughout this research. My dear friend Joseph Phillips for exposing me to the philosophy of Georges Battaille which I would end up incorporating here. Dr. Frank Edwards at the Rutgers Newark School of Criminal Justice for inspiring me and providing me with practical experience in thinking about econometric modelling and statistics. Dr. Jason Barr of the Rutgers Newark Economics Department for his curiosity and support as well. Michael Renne of the Math Department at Rutgers Newark for his support. All those science youtubers that make up what is colloquially known as the "University of Youtube", but specifically the channels Veritasium, 3blueOneBrown, DIBEOS, 2swap, CompuFlair, Optozorax, Abide By Reason, C to Y, MAKit, DataMListic, Mutual Information, and Welch Labs. If you are a science communicator, an honorary acknowledgement to you too!

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Note on Formalism:

This paper makes no ontological claims regarding the physical interpretation of quantum mechanics, such as the Many Worlds hypothesis. I subscribe to the "shut up and calculate" school of thought: the mathematical formalism of quantum mechanics is utilized here strictly as a set of statistical tools. Viewing these methods as instrumental rather than representational is necessary to maintain a focus on useful science rather than metaphysics.

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5. STOCHASTIC THERMODYNAMICS

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7. SOCIO-ECONOMIC DYNAMICS & ECONOPHYSICS

The Bataillean Framework (General Economy)

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- ***Note: This provides the philosophical basis for the "Housekeeping vs. Excess" decomposition.***

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Stoekl, A. (2007). *Bataille's Peak: Energy, Religion, and Postsustainability* (NED-New edition). University of Minnesota Press. http://www.jstor.org/stable/10.5749/j.ctttv59s

- ***Note: Essential secondary text relating Bataille's theories specifically to thermodynamics and energy expenditure.***

Fokker-Planck in Socio-Economics

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- ***Note: The core mathematical justification.***

Toscani, G. (2006). "Kinetic Models of Opinion Formation." *Communications in Mathematical Sciences*, 4(3), 481-496.

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Yakovenko, V.M. and Rosser, J.B. (2009). "Colloquium: Statistical Mechanics of Money, Wealth, and Income." *Reviews of Modern Physics*, 81(4), 1703-1725. DOI: 10.1103/RevModPhys.81.1703

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- ***Note: Applies Field Theory/Gauge Theory explicitly to pricing and inflation dynamics.***

Mantegna, R.N. and Stanley, H.E. (2000). *Introduction to Econophysics: Correlations and Complexity in Finance*. Cambridge University Press.

8. QUANTUM MECHANICS

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- ***Note: It was Nelson who developed the score based Osmotic Velocity here, along with the realization that diffusion could be used as an effective quantum constant.***

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The Hydrodynamic Formulation (Madelung)

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- ***Note: The origin of the decomposition*** $\psi = \sqrt{\rho}e^{iS/\hbar}$ ***and the fluid interpretation.***

Bohm, D. (1952). "A Suggested Interpretation of the Quantum Theory in Terms of 'Hidden' Variables. I & II." *Physical Review*, 85(2), 166-193. DOI: 10.1103/PhysRev.85.166.

- ***Note: This text rigorously formalized the 'Quantum Potential' (*** Q ***) used in the analysis.***

The Wave Equation

Schrödinger, E. (1926). "Quantisierung als Eigenwertproblem" (Quantization as an Eigenvalue Problem). *Annalen der Physik*, 384(4), 361-376. DOI: 10.1002/andp.19263840404.

Uncertainty & Kinematics

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- ***Note: The source of the kinematic inequality*** $\Delta x \Delta p \geq \hbar/2$ ***observed in the Market Phase Space***

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